# Chapter 1

# Introduction

## 1.1 Background and Motivation

Financial risks develop through events that affect firms and the financial environment in which they operate and information about the risk reaches investors gradually, often before its economic consequences can be fully assessed.Articles document incidents as they occur and document their developments. As new information becomes available, the volume and content of reporting change.

These changes can be examined using quantitative methods for example: the number of articles associated with a risk provides a measure of the attention it receives, the language used in those articles offers further information about how the risk is being discussed.

This thesis studies five categories of emerging risk: Cyber Risk, Climate Risk, Pandemic Risk, Supply Chain Risk, and AI-Related Risk. The analysis uses newspaper articles published between 2014 and 2023 to build indicators that describe the evolution of reporting associated with each category.

The study begins with the measurement of these risks across the news corpus. Then it examines the behavior of the resulting indicators over time. The final part focuses on Cyber Risk and investigates whether changes in cyber-related news contain information about subsequent equity returns.

The financial application is inspired on the work of **cherubini_cyber_risk<empty citation>**, who examine the relationship between a text-based measure of cyber-risk attention and stock market returns. Their approach provides a starting point for investigating the market relevance of the indicators developed in this thesis.

## 1.2 Research Problem

Emerging risks are difficult to measure consistently because the information available about them changes as events unfold.

The initial problem concerns the identification of relevant news. The newspaper database contains articles covering many subjects, and the same risk can be described using different expressions.

A classification procedure is needed to identify reporting associated with each of the five categories.

The next problem concerns the measurement of changes in reporting. Consider two periods in which the number of cyber-related articles increases. In the first, newspapers may repeatedly cover an incident that has already been described. In the second, the increase may involve reports of previously unknown attacks affecting several firms. Both periods produce higher article counts, although the content of the incoming information differs. This motivates the construction of indicators that track the distinctiveness of new articles and changes in the language associated with each risk category.

The temporal dimension introduces a further question. A risk may receive a relatively high share of coverage for several months, while another may experience a sudden increase after a period of limited attention. Identifying these episodes requires comparing current reporting with the previous history of the same theme.

The thesis addresses these measurement problems through four indicators: Coverage, Novelty, Semantic Drift and Burst Detection.

The final research problem concerns the financial information contained in these measures. Once the indicators have been constructed, their relationship with market outcomes can be investigated. The empirical application focuses on Cyber Risk and examines whether changes in news-based indicators are associated with subsequent weekly equity portfolio excess returns.

## 1.3 Research Objectives

The first objective is to make sure to work with a news database suitable for the measurement of emerging risks. The material consists of articles published between 2014 and 2023 by five newspapers: the Financial Times, Wall Street Journal, New York Times, Los Angeles Times, and USA Today. After validation and deduplication, the database contains 2,249,255 articles.

The second objective is to identify reporting associated with the five selected risk categories. The classification procedure combines lexical rules with supervised text classification. Manually reviewed observations are used to evaluate the ability of the models to identify relevant articles. The resulting thematic scores provide the basis for the subsequent construction of the indicators.

The third objective is to represent the semantic content of the articles numerically. Document embeddings allow articles to be compared according to their textual content. These representations are used to examine how incoming news differs from earlier reporting and how the content associated with a risk category changes over time.

The fourth objective is to construct four measures of emerging risk from classified articles and their semantic representations. Here four are constructed: Coverage, Novelty, Semantic Drift, Burst Detection.
The indicators are constructed across the five risk categories. Their empirical analysis examines how

the measures evolve during 2015–2023.

The fifth objective is to investigate Cyber Risk in details to assess the market relevance of the cyber-risk indicators. Following the approach used by **cherubini_cyber_risk<empty citation>**, the thesis estimates models linking changes in cyber-related news measures to weekly equity excess returns.

The final analysis considers two equally weighted portfolios: the Diversified portfolio contains 195 firms from several industries and the Attacked + Security portfolio contains 76 firms, including 66 companies classified as having experienced cyberattacks and 10 cybersecurity firms.
The empirical models examine the relationship between lagged changes in the indicators and subsequent returns. An out-of-sample forecasting exercise then evaluates whether Coverage improves one-week-ahead return predictions relative to an autoregressive benchmark. Additional tests investigate the estimated relationships during periods of unusually intense cyber-risk reporting.

## 1.4 Contributions

The first contribution of the thesis is a common measurement framework for five emerging risk categories. The framework combines information about the frequency of reporting with information extracted from article content. Novelty describes how different incoming articles are from earlier news. Semantic Drift follows changes in the representation of each theme. Coverage measures the share of attention received by a risk, while Burst Detection identifies episodes of unusually strong increases in that attention. The empirical analysis shows how these measures behave across the five categories. Their time series reveal differences in the timing of changes in coverage and semantic content. The indicators also allow periods of intense reporting to be examined in relation to the information published during those periods.

The second contribution concerns the analysis of Cyber Risk. The thesis analizes the evolution of its indicators and identifies episodes in which changes in attention coincide with changes in semantic content. The analysis also documents periods in which the indicators follow different patterns.

The third contribution is the investigation of the financial relevance of cyber-risk news indicators. Building on the approach of Cherubini et al., the thesis examines the relationship between the constructed indicators and weekly equity returns for two portfolios with different compositions.

The results indicate that Coverage has the clearest statistical association with excess returns. Its six lagged changes are significant in both portfolios, including combinations with Novelty and/or Semantic Drift.
Adding Coverage increases adjusted $R^2$ by 3.64 percentage points for the Diversified portfolio and 4.43 percentage points for the Attacked + Security portfolio. The out-of-sample exercise also produces lower one-week-ahead root mean squared errors, with reductions of 1.19% and 1.61%, respectively, relative to the autoregressive benchmark.
The analysis of Burst episodes examines whether unusually intense cyber-risk attention changes the estimated relationship between news indicators and returns. The tests do not identify a statistically significant difference during these periods. The joint significance of Coverage remains when contem-

poraneous Burst weeks are excluded from the return regressions.

The financial results document a predictive association between cyber-risk Coverage and weekly excess returns within the portfolios and sample periods examined.

## 1.5 Structure of the Thesis

Chapter 2 reviews the literature studied before writing this thesis and the theoretical methods used in the financial analysis.
Chapter 3 describes the construction of the newspaper database.
Chapter 4 develops the thematic classification framework used to identify articles associated with the five risk categories. It also reports the evaluation of the classification procedure.
Chapter 5 presents the construction of document embeddings and explains how semantic representations are used to compare articles.
Chapter 6 develops the four emerging-risk indicators.
Chapter 7 examines the behavior of these indicators across the five risk categories during the sample period. It then focuses more closely on Cyber Risk, considering the evolution of its measures and the episodes identified in the news.
Chapter 8 investigates the market relevance of cyber-risk news. It estimates the dynamic relationship between the indicators and weekly equity excess returns, evaluates the forecasting contribution of Coverage, and examines the results during Burst episodes.
Chapter 9 contains the conclusions of the thesis.

# Chapter 2

# Literature Review and Theoretical Foundations

This thesis uses information from financial news to measure five risk themes and examine the relationship between cyber-risk news and weekly equity returns. The methods connect language representations to financial prices and use statistical tools for text classification and time-series inference. This chapter develops the theoretical foundations for Chapters 4–8.

## 2.1 Financial News in the Literature

### 2.1.1 Language representations and expected returns

Chen, Kelly, and Xiu study whether embeddings extracted from financial news by large language models help predict cross-sectional stock returns **chen_kelly_xiu**. Their analysis examines the return-predictive information contained in language-model representations of financial news. The study serves as a benchmark on using news text as a numerical signal source in empirical asset pricing. An embedding is a numerical representation of a text where the coordinates are learned from language data. This representation makes it possible for a predictive model to learn patterns from all over the article, including relationships between phrases which have little in common in terms of vocabulary. Chen, Kelly, and Xiu leverage this property to establish the link between the content of news articles and their expected returns. This thesis uses document embeddings to compare an article with previous articles and observe the evolution of the content related to a risk theme. The embeddings enter the construction of Novelty and Semantic Drift in Chapter 6.

Didisheim, Kelly, Pourmohammadi, and Tian examine how predictable news content affects estimates of market responses to information **didisheim_inefficient_pricing**. They predict news embeddings using prevailing stock characteristics and study the residual component. In simplified notation, their distinction can be expressed as

$$\mathbf{e}_{i,t} = \widehat{\mathbf{e}}_{i,t} + \mathbf{u}_{i,t}, \tag{2.1}$$

where $\widehat{\mathbf{e}}_{i,t}$ is the predicted representation and $\mathbf{u}_{i,t}$ captures the portion left unexplained by that prediction. The authors refer to this residual information as a news shock. Their results examine how the component of news that is unexpected given observable stock characteristics relates to subsequent returns.

Their analysis gives a precise interpretation to an informational surprise: it depends on what could have been predicted from the available conditioning variables. Chapter 6 measures semantic distinctiveness by comparing each article with earlier reporting. The historical comparison defines information relative to earlier articles. Didisheim et al. instead construct residual shocks relative to a prediction based on stock characteristics.

### 2.1.2 Cyber-risk news and equity returns

Cherubini, Della Lunga, Figà Talamanca, and Guardabascio construct a text-based index of cyber-risk attention from newspaper coverage and investigate its dynamic relationship with US sector-level equity returns **cherubini_cyber_risk**. Their analysis uses weekly observations and a vector autoregressive framework. Their empirical framework estimates the dynamic relationship between changes in cyber-risk news and equity returns across sectors.
Their paper motivates the market-relevance analysis in Chapter 8. Both studies use newspaper-based measures and weekly financial data, with six lags in the dynamic specification. The thesis examines Coverage alongside Novelty and Semantic Drift and tests their association with two equally weighted portfolios. Its return equation uses excess returns in levels and lagged changes in the news indicators. The lag coefficients measure conditional associations between past news and current portfolio returns.

## 2.2 Statistical Foundations of Text Classification

### 2.2.1 Lexical rules and weak supervision

The classification problem in Chapter 4 assigns thematic relevance to articles for five categories: cyber risk, climate risk, pandemic risk, supply-chain risk, and AI-related risk. A keyword baseline searches for terms associated with each theme. The terms selected for each rule determine which articles are captured. An article may describe a cyber incident without using a selected keyword; the same keyword may also appear in an unrelated discussion.
Weak supervision uses labeling rules to generate training information from text. A labeling function $\lambda_j(x_i)$ assigns evidence about article $i$ from a specified textual pattern or contextual condition. Different functions may cover different parts of the text and can disagree on the same article. The resulting labels require validation against manually assessed observations.

### 2.2.2 TF–IDF and logistic regression

Term Frequency–Inverse Document Frequency (TF–IDF) maps documents to sparse vectors. A term receives greater weight when it appears frequently in a document and less weight when it is common throughout the corpus. A conventional expression is

$$\text{tfidf}(w, d) = \text{tf}(w, d)\ \text{idf}(w), \tag{2.2}$$

where $w$ denotes a term and $d$ a document. The thesis uses unigram and bigram features in its supervised thematic classifier. These features represent word occurrences and adjacent word pairs, allowing the model to associate expressions with particular risk categories.

Logistic regression relates a feature vector $\mathbf{x}_i$ to a binary label through the logistic link:

$$\Pr(Y_i = 1 \mid \mathbf{x}_i) = \frac{1}{1 + \exp[-(\beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta})]}. \tag{2.3}$$

The fitted coefficients determine how textual features contribute to the classifier score. With many sparse predictors, $L_2$ regularization penalizes large coefficients and limits sensitivity to the training sample. Class weighting changes the contribution of positive and negative examples to the fitting objective when their frequencies differ. Chapter 4 applies both choices.

Classification performance is assessed from true positives ($TP$), false positives ($FP$), and false negatives ($FN$). Precision measures the share of predicted positives that are correct, while recall measures the share of actual positives recovered. Their harmonic mean is the $F_1$ score:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2\ \text{Precision}\ \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2.4}$$

Precision falls when the classifier assigns a risk label to unrelated articles; recall falls when it misses relevant articles. The supervised model also produces continuous thematic scores. Their interpretation depends on model training and calibration, so Chapter 4 treats them as relevance scores rather than calibrated probabilities of true theme membership.

## 2.3 Semantic Representations of Financial News

### 2.3.1 Document embeddings

Document embeddings transform an article into a vector $\mathbf{e}_i \in \mathbb{R}^d$ to produce contextual representations by processing the words in a document together. Their vector geometry makes comparisons between articles that describe similar events using different vocabulary possible. Chapter 5 uses the model `all-MiniLM-L6-v2`, which produces 384-dimensional document representations.

The embedding space provides a common numerical representation for comparisons across articles and publication dates. An article describing a ransomware attack, for example, may be close to another account of a network intrusion even if the two articles use different terms. This property makes embeddings suitable for studying how the content of risk-related reporting evolves.

### 2.3.2 Cosine similarity and temporal comparison

Cosine similarity measures the angle between two embedding vectors:

$$\text{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|\,\|\mathbf{e}_j\|}. \tag{2.5}$$

For vectors normalized to unit length, cosine similarity equals their inner product. Values closer to one indicate greater directional similarity. The resulting score compares vector directions in the embedding space.

Temporal comparisons require a defined historical information set. A document published in week $t$ can be compared with articles available before that week; a monthly thematic representation can likewise be compared with a historical reference. These comparisons provide the geometric basis for

the semantic measures developed in Chapter 6. The definitions of Novelty and Semantic Drift, together with the construction of Coverage and Burst Detection, are given there.

## 2.4 Econometric Models and Statistical Inference

### 2.4.1 Autoregressive models and VAR specifications

The financial analysis studies weekly portfolio excess returns $R_{P,t}$ and the weekly changes in news indicators $\Delta I_t = I_t - I_{t-1}$. An autoregressive model uses lagged values of a series to account for its own time dependence:

$$R_{P,t} = a_P + \sum_{\ell=1}^{p} b_{P,\ell} R_{P,t-\ell} + \varepsilon_{P,t}. \tag{2.6}$$

An AR(6) model serves as the forecasting benchmark in Chapter 8. The augmented specification adds six lags of the change in cyber-risk Coverage.
A vector autoregression (VAR) models several series jointly. For a vector $\mathbf{Y}_t$, a VAR($p$) is

$$\mathbf{Y}_t = \mathbf{a} + \sum_{\ell=1}^{p} A_\ell \mathbf{Y}_{t-\ell} + \mathbf{u}_t. \tag{2.7}$$

The thesis estimates bivariate VAR(6) specifications with $\mathbf{Y}_{P,t} = (R_{P,t}, \Delta I_t)^\top$. The return equation is

$$R_{P,t} = a_P + \sum_{\ell=1}^{6} b_{P,\ell} R_{P,t-\ell} + \sum_{\ell=1}^{6} c_{P,\ell} \Delta I_{t-\ell} + \varepsilon_{P,t}, \tag{2.8}$$

where the coefficient $c_{P,\ell}$ measures the association between a change in the indicator $\ell$ weeks earlier and the current excess return, conditional on the other regressors. Models containing several indicators include a separate set of six lags for each measure. The equations are estimated by ordinary least squares (OLS), which minimizes the sum of squared residuals. For a regression $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, the OLS estimator is $\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$ when the inverse exists.

### 2.4.2 Wald tests and HAC covariance estimation

A Wald test evaluates linear restrictions on estimated coefficients. In general form, the null hypothesis is $H_0 : R\boldsymbol{\beta} = \mathbf{r}$. Given an estimated covariance matrix $\widehat{V}$, the test statistic is

$$W = (R\widehat{\boldsymbol{\beta}} - \mathbf{r})^\top (R\widehat{V}R^\top)^{-1} (R\widehat{\boldsymbol{\beta}} - \mathbf{r}). \tag{2.9}$$

Under standard regularity conditions, $W$ converges under the null to a chi-squared distribution with degrees of freedom equal to the number of independent restrictions. In the return equation, the individual-indicator null is $H_0 : c_{P,1} = \cdots = c_{P,6} = 0$. Combined models allow the joint significance of several lag blocks to be tested. Conditional tests retain the other news indicators while restricting one block of coefficients.

Weekly regression errors may exhibit changing variance and serial dependence. Heteroskedasticity- and autocorrelation-consistent (HAC) covariance estimation accounts for these features when computing

coefficient uncertainty. A generic long-run covariance estimate combines contemporaneous covariance with weighted lagged autocovariances:

$$\widehat{\Omega} = \widehat{\Gamma}_0 + \sum_{\ell=1}^{L} w_\ell (\widehat{\Gamma}_\ell + \widehat{\Gamma}_\ell^\top). \tag{2.10}$$

The bandwidth $L$ determines the number of included autocovariance lags. Chapter 8 uses HAC(6) for its main Wald tests and HAC(12) as a bandwidth sensitivity check. HAC changes the estimated coefficient covariance matrix and the resulting test statistics. The OLS point estimates remain unchanged. A small Wald-test $p$-value indicates that the data are inconsistent with the imposed restrictions under the chosen inferential framework.

## 2.5 Forecast Evaluation and Conditional Relationships

### 2.5.1 Expanding-window forecasts and nested-model comparison

An expanding-window exercise estimates the forecasting model using observations available up to each forecast origin. The next observation is predicted, the sample is extended, and the procedure is repeated. This design evaluates the information available to the model at the time each forecast is formed.
Chapter 8 compares one-week-ahead forecasts from an AR(6) benchmark with forecasts from an augmented model containing six lagged changes in Coverage. Let $e_{0,t}$ and $e_{1,t}$ be the corresponding forecast errors. Root mean squared error (RMSE) and out-of-sample $R^2$ are

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_{j,t}^2}, \qquad R^2_{\mathrm{OOS}} = 1 - \frac{\sum_t e_{1,t}^2}{\sum_t e_{0,t}^2}. \tag{2.11}$$

A positive out-of-sample $R^2$ indicates lower squared forecast error for the augmented model over the evaluation period.
The Clark–West test addresses comparisons between nested forecasting models. Estimating additional coefficients can increase the larger model's squared forecast errors under the null of no incremental predictive content. The test adjusts the loss differential for the difference between the two forecasts. If $\widehat{R}_{0,t}$ and $\widehat{R}_{1,t}$ denote the benchmark and augmented forecasts, the adjusted differential is

$$f_t = e_{0,t}^2 - e_{1,t}^2 + (\widehat{R}_{0,t} - \widehat{R}_{1,t})^2. \tag{2.12}$$

The one-tailed test on its mean assesses the improvement in forecasting accuracy made by the enhanced model after the nesting correction. The thesis uses HAC(6) tests on this difference. This test allows an inferential assessment of the forecasting variable in addition to the reported RMSEs.

### 2.5.2 Burst interactions and sensitivity checks

Interaction regressions check whether a fitted relationship depends on the state. Let $S_{I,t}$ represent lagged changes in a news indicator, and let $B_{t-1}$ represent a Burst state at time *t-1*. A general specification is

$$R_{P,t} = \alpha_P + \beta_P S_{I,t} + \gamma_P B_{t-1} + \delta_P S_{I,t} B_{t-1} + \mathbf{z}_t^\top \boldsymbol{\theta}_P + \varepsilon_{P,t}, \tag{2.13}$$

where $\mathbf{z}_t$ collects the other controls. The slope on $S_{I,t}$ is $\beta_P$ in ordinary weeks and $\beta_P + \delta_P$ following a Burst state. Testing $H_0 : \delta_P = 0$ assesses whether the two slopes differ. Chapter 8 uses HAC(6) inference for these regressions and examines the sensitivity of its Coverage results to the exclusion of contemporaneous Burst weeks.

Chapter 8 applies these tests to weekly portfolio returns and reports the out-of-sample comparison between the AR(6) benchmark and the model augmented with lagged Coverage.